LESSON PLAN

ENGINEERING METROLOGY (10)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Standard maintenance procedure, inspection, standard maintenance and safety precaution
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Standard maintenance procedure, inspection, standard maintenance and safety precaution
5. Specific Objectives	10.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of standard maintenance checklist usage, inspection of calibrated tools, and verification of safety precaution compliance.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of standard maintenance procedures?
(b) List three common safety precautions required in an engineering workshop.
(c) Why is regular inspection of metrology equipment essential?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of standard maintenance procedure • Steps for routine equipment inspection • Importance of calibration records • Identification of wear and tear signs • Documentation of maintenance activities	L & Ds	PPT, IB	15 min
(b)	• Safety precautions during tool handling • Proper storage of metrology instruments • Emergency response protocols in workshop • Personal protective equipment requirements • Handling hazardous materials safely	L & Ds	PPT, PDF	20 min
(c)	• Conducting a mock inspection exercise • Reviewing standard maintenance logs • Identifying non-compliance issues • Corrective actions for safety violations • Final verification of maintenance standards	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between routine inspection and standard maintenance.
(b) Describe the correct procedure for storing precision metrology tools.
(c) What steps should be taken if a safety precaution is violated during maintenance?

12. Home Assignment and Follow Up	Prepare a brief report on the standard maintenance procedure for the next lesson plan (LP 02), focusing on specific equipment used in your unit.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Instructor, P No. 12345	Senior Instructor, Senior Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min